The derivative of e^x — 0.3.9

Goal: $\frac{d}{dx} e^x = e^x$.

Recall: we showed $\lim_{n\to\infty} A\left(1+\frac{r}{n}\right)^n = \cdots = Ae^r$

$\therefore \quad e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n$

<u>Fact</u>: $(a+b)^2 = a^2+2ab+b^2$

$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$

$(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+b^4$

$$\begin{array}{c} 1 \\ 1 \quad 1 \\ 1 \quad 2 \quad 1 \\ 1 \quad 3 \quad 3 \quad 1 \\ 1 \quad 4 \quad 6 \quad 4 \quad 1 \\ 1 \quad 5 \quad 10 \quad 10 \quad 5 \quad 1 \end{array}$$

$$\left(1+\frac{h}{n}\right)^n = 1^n + n\cdot 1^{n-1}\cdot\frac{h}{n} + h^2(\text{stuff})$$

$$= 1 + h + h^2(\text{stuff})$$

Intermediate goal: show that $f'(0) = 1$ for $f(x) = e^x$

That is, want to show that $\lim_{h \to 0} \dfrac{e^h - e^0}{h - 0} = 1$

$$\rightarrow \lim_{h \to 0} \frac{e^h - e^0}{h - 0} = \lim_{h \to 0} \frac{e^h - 1}{h} = \lim_{h \to 0} \frac{1}{h} (e^h - 1)$$

$$= \lim_{h \to 0} \frac{1}{h} \left(\lim_{n \to \infty} \left(1 + \frac{h}{n}\right)^n - 1 \right)$$

$$= \lim_{h \to 0} \frac{1}{h} \left(\lim_{n \to \infty} (1 + h + h^2 \cdot \text{stuff}) - 1 \right)$$

$$= \lim_{h \to 0} \frac{1}{h} (\underline{1} + h + h^2 \cdot \text{stuff} \underline{- 1}) = \lim_{h \to 0} \frac{1}{h} (h + h^2 \cdot \text{stuff})$$

$$= \lim_{h \to 0} 1 + h \cdot \text{stuff} = 1 + 0 \cdot \text{stuff} = 1. \quad \square$$

$$\frac{d}{dx} e^x = \lim_{h \to 0} \frac{e^{x+h} - e^x}{h} = \lim_{h \to 0} \frac{e^x e^h - e^x}{h}$$

$$= e^x \lim_{h \to 0} \frac{e^h - 1}{h} = e^x \cdot 1 = e^x.$$